Supplementary Material (Online Resource 1): Exact stationary majority fraction in group-flip consensus dynamics on quenched hypergraph balls

Pham Le Dat Nguyen[1*]

[1*]Independent Researcher, Ha Noi, Viet Nam.

Corresponding author(s). E-mail(s): phmldtngyn.777@gmail.com;

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Notation and conventions

Objects

$H = (V, E)$, $|V| = n$; hypergraph distance $d(\cdot, \cdot)$ is the minimum number of hyperedge steps (a step from i reaches any node sharing a hyperedge with i); d is symmetric. Ball $B_i = B_i(r) = \{j : d(i, j) \le r\}$, $\beta_i = |B_i|$, and β when all balls have the same size; coverage $q = \beta/n$; residual $R_i = V \setminus B_i$. Conditions: **R1** $\beta_i > n/2$ for all i; **R2** $\mathrm{diam}_H(R_i) \le r$ for all i. Dynamics: each node fires at rate 1; a fire at i moves all nodes of colour σ_i inside B_i to a target colour drawn uniformly among the other $Q-1$ colours. Observables: stationary majority fraction M; number K of fires between consecutive rotations (changes of the majority label) and physical cycle time T. Containment observables c_G and $\tilde{c}$ as in Eqs. (2)–(3) of the main text, with $\tilde{c} = 1$ if all residuals are empty.

Status of statements

Theorem, *Proposition*, *Lemma*, *Corollary*: a complete proof is given here. *Claim*: a statement with a proof sketch only; it is not used as a hypothesis of any theorem.

Conjecture: supported by numerical evidence, no proof. *Heuristic*: structural statement without a proof attempt. *Protocol*: a measurement or statistical design. *Note*: methodological record. *Remark*: scope clarification.

Correspondence with the main text

Main-text Theorem 1 is Proposition C.1; Lemmas 1–2 are Lemmas C.2–C.3; Theorem 2 is Theorem C.6; Propositions 1–2 are Propositions D.3 and D.1; Propositions 3–4 are Propositions E.1–E.3; Conjecture 1 is Conjecture G.1; Protocol 1 is Protocol F.4. All exact formulas use $q = \beta/n$.

A Well-mixed limit

Lemma A.1 (Count-chain reduction) *If $B_i = V$ for all i, the colour-count vector $x = (x_1, \ldots, x_Q)$, $\sum_k x_k = n$, is a Markov chain on the simplex with transitions $x \to x + x_k(e_l - e_k)$ at rate $x_k/(Q-1)$ for each ordered pair (k, l), $k \neq l$.*

Proof A fire at a node of colour k (total rate x_k) moves the entire class k (since $B_i = V$) to the target l, chosen with probability $1/(Q-1)$. The transition rates depend on the configuration only through x, so the lumping onto counts is exact. □

Lemma A.2 (Pure death of live colours) *Let L_t be the number of colours with at least one node. Then L_t is a pure-death chain on $\{1, \ldots, Q\}$ with $L \to L-1$ at rate $n(L-1)/(Q-1)$; $L = 1$ is absorbing.*

Proof Every node carries a live colour, so $\sum_{k\ \mathrm{live}} x_k = n$ at all times. A fire from live class k decreases L if and only if the target is another live class, which has probability $(L-1)/(Q-1)$; summing over live k gives rate $n(L-1)/(Q-1)$. A target that is an absent colour only relabels the firing class and leaves L unchanged; no event increases L. At $L = 1$ the rate is 0. □

Theorem A.3 (First-consensus time) *If L_0 colours are initially live, then, with node firing rate 1,*

$$E[T_{\mathrm{cons}}] = \frac{Q-1}{n} \sum_{\ell=2}^{L_0} \frac{1}{\ell - 1} = \frac{Q-1}{n} H_{L_0 - 1}.$$

In particular $E[T_{\mathrm{cons}}] = \frac{Q-1}{n} H_{Q-1}$ when $L_0 = Q$, and $E[T_{\mathrm{cons}}] = 1/n$ for $Q = 2$.

Proof By Lemma A.2 the holding time at level ℓ is exponential with rate $n(\ell-1)/(Q-1)$; sum the means over $\ell = L_0, \ldots, 2$. For $Q = 2$ the first fire always merges the two classes. □

Remark A.3.1 The harmonic form is specific to whole-class moves. Updates that sample pairs of nodes lead to different prefactors and are not considered here.

Theorem A.4 (Minority advantage, $Q = 2$) *From $(a, n-a)$, the class of size a wins the first consensus with probability $1 - a/n$.*

Proof With $Q = 2$ the first fire decides. If the firing node is in the class of size a (probability a/n), that class moves into the other colour and loses; otherwise it wins. □

Theorem A.5 (Three colours, one absent) *Let $Q = 3$, $a, b > 0$ with $a + b = n$, colours 1 and 2 live with sizes a and b, and colour 3 absent. With $D = a^2 + 3ab + b^2$, the probability that colour k wins the first consensus is*

$$P_1 = \frac{b(a+b)}{D}, \qquad P_2 = \frac{a(a+b)}{D}, \qquad P_3 = \frac{ab}{D},$$

and $P_1 + P_2 + P_3 = 1$. Here P_3 is the probability that the absent colour is regenerated and wins.

Proof First-step analysis with $F = P_1(a, b \text{ live})$, $G = P_1(b, a \text{ live})$ (roles exchanged), and $U = P_1(\text{colour 1 absent, live } a, b)$; the total rate is $a + b$:

$$\begin{aligned}
(a+b)F &= a\Big[\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}U\Big] + b\Big[\tfrac{1}{2}\cdot 1 + \tfrac{1}{2}F\Big],\\
(a+b)U &= a\Big[\tfrac{1}{2}F + \tfrac{1}{2}\cdot 0\Big] + b\Big[\tfrac{1}{2}G + \tfrac{1}{2}\cdot 0\Big],\\
(a+b)G &= b\Big[\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}U\Big] + a\Big[\tfrac{1}{2}\cdot 1 + \tfrac{1}{2}G\Big].
\end{aligned}$$

From the first and third equations, $F = (aU + b)/(2a + b)$ and $G = (bU + a)/(2b + a)$. Substituting into the second gives

$$U\Big[2(a+b) - \frac{a^2}{2a+b} - \frac{b^2}{2b+a}\Big] = ab\Big[\frac{1}{2a+b} + \frac{1}{2b+a}\Big].$$

With $t = a + b$ and $p = ab$, $(2a + b)(2b + a) = 2t^2 + p$; the bracket equals $3t(t^2 + p)/(2t^2 + p)$ and the right-hand side $3pt/(2t^2 + p)$, so $U = p/(t^2 + p) = ab/D$, since $t^2 + p = D$. Back-substitution and $2a^2 + 3ab + b^2 = (2a + b)(a + b)$ give $F = b(a + b)/D$ and $G = a(a + b)/D$. Finally $(a + b)^2 + ab = D$. □

Remark A.5.1 For $a = b$: $P_1 = P_2 = 2/5$ and $P_3 = 1/5$. For $a < b$: $P_1 > P_2$, so the smaller class keeps its advantage. As $b \to 0^+$ the limits of Theorem A.4 are recovered.

Theorem A.6 (General Q) *Consensus states are absorbing and every non-consensus state is transient (Lemma A.2). The winning probabilities are the unique solution of the first-step system $(I - P_{TT})x = P_{TA}x_A$ over the transient states T, where P_{TT} is substochastic with spectral radius < 1, so $I - P_{TT}$ is invertible.*

Proof Standard hitting-probability theory for finite absorbing chains; invertibility follows from transience (Neumann series). □

Remark A.6.1 No closed form is claimed for $Q \geq 4$.

B Coverage and ball geometry

Proposition B.1 (Ball sizes, no wrap-around) *On the d-dimensional hypergrid whose hyperedges are three consecutive nodes along one axis, one hyperedge step moves at most two lattice sites along one axis. The cost of an offset o along an axis is $\lceil |o|/2 \rceil$, and as long as balls do not wrap around, $|B_r| = \sum_{c_1+\cdots+c_d \le r} \prod_k w(c_k)$ with $w(0) = 1$ and $w(c) = 4$ for $c \ge 1$. Hence $|B_r| = 4r + 1$ in 1D and $|B_r| = 8r^2 + 1$ in 2D.*

Proof The offsets of cost $c \ge 1$ along one axis are $\pm(2c-1)$ and $\pm 2c$, four in total; the count follows by convolution over axes. In 2D, $(1 + 4r) + \sum_{c=1}^{r} 4(1 + 4(r-c)) = 8r^2 + 1$. □

Proposition B.2 (1D: distance, ball size, R1, R2) *On the 1D hypergrid, $d(i,j) = \lceil \delta(i,j)/2 \rceil$ with δ the circular lattice distance, and $\beta = \min(n, 4r+1)$. Exactly (for every finite n): R1 $\iff$ $2\beta > n$, and R2 $\iff$ $\beta = n$ or $3\beta \ge 2n - 1$; when $\beta = 4r + 1 < n$ the latter is equivalent to $n \le 6r + 2$.*

Proof A hyperedge step moves ± 1 or ± 2 sites, which gives the distance formula and shows that balls are arcs. R1 is $\beta > n/2$. For R2 with $\beta < n$: the residual is an arc of $n - \beta$ nodes, of lattice diameter $n - \beta - 1$ and hypergraph diameter $\lceil (n - \beta - 1)/2 \rceil$; R2 is $\lceil (n - \beta - 1)/2 \rceil \le r$, i.e. $n - \beta - 1 \le 2r$; with $r = (\beta - 1)/4$ this is $3\beta \ge 2n - 1$, and substituting $\beta = 4r + 1$ gives $n \le 6r + 2$. Since R1 gives $\beta > n/2$, the residual arc has length $n - \beta < n/2$, strictly less than half the circle; its hypergraph diameter is therefore computed along the arc itself, without wrap-around. □

Proposition B.3 (Coverage radius, no wrap-around) *For fixed $0 < q < 1$ and d, the smallest radius with $|B_r| \ge qn$ in the no-wrap regime satisfies $r_{\min} = \Theta(n^{1/d})$, with constants from Proposition B.1.*

Proof $|B_r|$ is a strictly increasing polynomial of degree d in r, so the radius is fixed between consecutive integers. □

Claim B.4 (Wrapped regime; proof sketch only) *For fixed q on d-dimensional tori, $|B_r|/n$ converges to the rescaled volume of the corresponding diamond, with $O(n^{-1/d})$ boundary corrections. The argument (lattice counts as Riemann sums) is sketched, not proved, and has been checked numerically only. The claim is not used in any result below.*

Claim B.5 (Sparse configuration model; proof sketch only) *If $(k-1)^{2r+1} = o(n)$, then for the configuration model (plain or conditioned to be simple) $|B_r|$ is asymptotically the non-backtracking tree volume, with collision probability $o(1)$ by a union bound over endpoint checks, each of order $(k-1)^{2r}/n$. The union bound is not written out. The claim motivates the truncated-ball design of Protocol F.4 and is not used in any result below.*

C Exact band at high coverage

Lemma C.0 (Entry and initial conditions) *Let $Q = 2$ and assume R1∧R2. (i) From a monochrome state, a fire at i creates the minority R_i. (ii) From any strict-majority state, the minority after the next rotation is contained in R_i, where i is the node whose fire produced it. (iii) Tie states are left at the first event and never re-entered. (iv) For any initial configuration (including i.i.d. colourings), the post-rotation (epoch) chain enters a closed communicating class in finite time almost surely. Different closed classes may support different stationary distributions; however, Lemma C.5 shows that the stationary mean minority fraction is the same on every closed class. Consequently, the stationary majority fraction in Theorem C.6 is independent of the initial law and of the closed class.*

Proof (i) The moved class occupies B_i; since $\beta_i > n/2$ it is the new majority and the minority is R_i. (ii) is Lemma C.2. (iii) is part of Proposition C.1. (iv) The epoch class is closed by Lemma C.2 and is reached at the first rotation or birth epoch, which occurs almost surely in finite time on a finite state space. Stationary distributions of a finite chain with several closed classes need not be unique; the class-independence used in Theorem C.6 does not follow from support on closed classes and is proved separately in Lemma C.5. □

Remark C.0.1 This lemma only justifies the use of stationary initial data for the observables of Theorem C.6; it does not assert uniqueness of the stationary distribution.

Proposition C.1 (Rotation identity; equal balls not required) *Let $Q = 2$ and $\beta_i > n/2$ for every i. From a strict-majority state with majority $\mathcal{M}$ and minority $\mathcal{A}$, a rotation occurs if and only if the firing node lies in $\mathcal{M}$; the instantaneous rotation rate is $|\mathcal{M}|$. Tie states are transient and carry zero stationary mass.*

Proof Majority fire at i: the class sizes after the fire are $|\mathcal{A}| + |\mathcal{M} \cap B_i|$ (new colour) and $|\mathcal{M} \setminus B_i|$ (old colour), and $|\mathcal{A}| + |\mathcal{M} \cap B_i| - |\mathcal{M} \setminus B_i| = (2\beta_i - n) + 2|\mathcal{A} \setminus B_i| > 0$ because $\beta_i > n/2$; the majority changes label. Minority fire at i: the class $\mathcal{M} \cup (\mathcal{A} \cap B_i)$ grows and $\mathcal{A} \setminus B_i$ shrinks, so the label is unchanged. Each node of $\mathcal{M}$ fires at rate 1, so the rotation rate is $|\mathcal{M}|$. Ties: with $|\mathcal{M}| = |\mathcal{A}|$ the same computation shows that any fire produces a strict majority, and strict-majority states map to strict-majority states; hence ties are transient. □

Remark C.1.1 In a majority fire only majority nodes inside B_i move; minority nodes inside B_i already carry the target colour and are unchanged.

Remark C.1.2 Only $\beta_i > n/2$ for the firing node is used. Equal balls first enter in Lemma C.3.

Lemma C.2 (Epoch invariant and self-containment) *Under R1∧R2 with $Q = 2$, every post-rotation minority A_m, created by a fire at i_m, satisfies $A_m \subseteq R_{i_m}$. Hence $A_m \subseteq B_j$ for all $j \in A_m$ (self-containment) and $|A_m| \le n - \beta_{i_m}$.*

Proof Base case: Lemma C.0(i). Induction: let $A \subseteq R_i$ be the current minority. R2 applied to R_i gives $R_i \subseteq B_j$ for every $j \in R_i$, hence $A \subseteq B_j$ for every $j \in A$. If the next fire is at $j \in A$, the whole minority moves and the system becomes monochrome; the following fire, at some i', creates $A' = R_{i'}$. If the next fire is at $i \in \mathcal{M}$, a rotation occurs (Proposition C.1) and the new minority is $\mathcal{M} \setminus B_i \subseteq R_i$. In both cases the new minority is contained in the residual of the node that created it, and self-containment follows again from R2. □

Lemma C.3 (Overlap identity; equal balls) *Under Lemma C.2 with $\beta_i \equiv \beta$, for i uniform in $\mathcal{M}$: $E[|A \cap B_i| \mid a] = a(\beta - a)/(n - a)$.*

Proof By symmetry of d, $j \in B_i \iff i \in B_j$, so $\sum_{i \in \mathcal{M}} |A \cap B_i| = \sum_{j \in A} |B_j \cap \mathcal{M}|$. For $j \in A$, self-containment gives $B_j \cap A = A$, hence $|B_j \cap \mathcal{M}| = \beta - a$. The sum is $a(\beta - a)$; divide by $|\mathcal{M}| = n - a$. □

Lemma C.4 (Affine conditional mean) *Under the assumptions of Lemma C.3, $E[a_{m+1} \mid a_m = a] = (1 - q)(n - a)$.*

Proof Minority fire (probability a/n): $a' = n - \beta$ (Lemma C.2). Majority fire (probability $1 - a/n$): $a' = n - \beta - a + |A \cap B_i|$, with conditional mean $n - \beta - a + a(\beta - a)/(n - a) = (n - \beta)(n - 2a)/(n - a)$. Weighting, $\frac{a}{n}(n - \beta) + \frac{n-a}{n} \cdot \frac{(n-\beta)(n-2a)}{n-a} = (n - \beta)\frac{n-a}{n} = (1 - q)(n - a)$. □

Lemma C.5 (Stationary epoch mean and cycle structure) *Let π be any invariant distribution of the epoch chain and $\bar{x} = E_\pi[a]/n$. Then $\bar{x} = (1 - q)/(2 - q)$, independently of π. Moreover $P(K = 2 \mid a) = a/n$, so $E_\pi[K] = 1 + \bar{x}$.*

Proof Taking π-expectations in Lemma C.4, $\bar{x}n = (1 - q)(n - \bar{x}n)$, whose unique solution is $\bar{x} = (1 - q)/(2 - q)$. This holds for every invariant π, in particular for one supported on any single closed class. A cycle contains a second fire if and only if its first fire is a minority fire, which has probability a/n given the epoch state. □

Theorem C.6 (Exact band curve) *Assume $Q = 2$, R1, R2, and equal balls β. Then $E[M] = (2 - q)/(3 - 2q)$.*

Proof Start the epoch chain in an invariant π (which exists, the state space being finite). Let $Y_m = \int M \, dt$ and τ_m be the reward and length of cycle m. The sequence (Y_m, τ_m) is stationary, as a functional of a stationary chain; it is not independent, and independence is not used. By the Birkhoff ergodic theorem, $\frac{1}{N}\sum_{m \le N} Y_m \to E[Y \mid \mathcal{I}]$ and $\frac{1}{N}\sum_{m \le N} \tau_m \to E[\tau \mid \mathcal{I}]$ almost surely, where $\mathcal{I}$ is the invariant σ-field; since $E[\tau \mid \mathcal{I}] \ge 1/n > 0$, the ratio converges almost surely to $E[Y \mid \mathcal{I}]/E[\tau \mid \mathcal{I}]$. On the event that the chain is in a given closed class c, these conditional expectations are the class-wise means $E_c[Y]$ and $E_c[\tau]$.

To compute them, note that a cycle starts immediately after a rotation and ends at the next one. Inter-event times are exponential with mean $1/n$ and independent of the colour

process. During the first interval the majority fraction is $1 - a_m/n$. If the first fire is a minority fire (probability a_m/n), the minority is erased, the system is monochrome during the second interval (majority fraction 1), and the next fire creates a new minority and a rotation, so $K = 2$; otherwise the first fire is a rotation and $K = 1$. Hence

$$E_c[Y] = \tfrac{1}{n} E_c[1 - a/n] + \tfrac{1}{n} P_c(K = 2) = \tfrac{1}{n}(1 - \bar{x}_c) + \tfrac{1}{n}\bar{x}_c = \tfrac{1}{n}$$

and $E_c[\tau] = \frac{1}{n}(1 + \bar{x}_c)$, where $\bar{x}_c$ is the mean under the invariant distribution of class c. Applying Lemma C.5 to the chain restricted to c gives $\bar{x}_c = (1 - q)/(2 - q)$ for every class. The ratio is therefore $1/(1 + \bar{x})$ on every class, hence almost surely overall, and substituting $\bar{x} = (1 - q)/(2 - q)$ gives $E[M] = (2 - q)/(3 - 2q)$. □

Corollary C.7 (Threshold location) *$f(q) = (2 - q)/(3 - 2q)$ is strictly increasing, with $f'(q) = (3 - 2q)^{-2} > 0$, $f(1/2) = 3/4$ and $f(2/3) = 4/5$. A crossing of $M = 3/4$ predicted by f would lie at $q = 1/2$, outside the exact regime. For 2D hypergrids the corollary applies only where R1∧R2 holds; at the finite sizes used here this was verified by breadth-first search ($L = 12$, $r = 4$, $\beta = 111$; $L = 30$, $r = 10$, $\beta = 699$). No 2D threshold theorem is claimed.*

Remark C.7.1 An operational threshold q_{dyn} defined by interpolating measured $M(q)$ at a level such as $3/4$ is a property of a measurement protocol, not a mathematical statement.

Theorem C.8 (Two-point cycle law) *Under the assumptions of Theorem C.6: $E[K] = 1 + \bar{x} = 1/E[M]$, $E[T] = (1 + \bar{x})/n$, and $\mathrm{CV}_T^2 = (7 - 8q + 2q^2)/(3 - 2q)^2$.*

Proof $K \in \{1, 2\}$ with $P(K = 2) = \bar{x}$ (Lemma C.5); the cycle length is $\tau = G_1 + BG_2$ with G_1, G_2 exponential of mean $1/n$ and B Bernoulli with mean $\bar{x}$. The G's are independent of the sequence of firing nodes, and B is determined by the firing node at the end of the first interval, hence independent of G_1 and G_2. Therefore $E[\tau] = (1 + \bar{x})/n$ and $\mathrm{Var}(\tau) = \frac{1}{n^2} + \frac{\bar{x}}{n^2} + \frac{\bar{x}(1-\bar{x})}{n^2} = \frac{1+2\bar{x}-\bar{x}^2}{n^2}$. With $\bar{x} = (1 - q)/(2 - q)$, $1 + \bar{x} = (3 - 2q)/(2 - q)$ and $1 + 2\bar{x} - \bar{x}^2 = (7 - 8q + 2q^2)/(2 - q)^2$, which gives CV_T^2. Finally $E[K] = 1 + \bar{x} = 1/E[M]$ by Theorem C.6. □

D Containment theory

Proposition D.1 (c_G on vertex-transitive and configuration geometries) *With i, j independent and uniform, $c_G = E_{i,j}\mathbf{1}\{R_i \subseteq B_j\}$. On a vertex-transitive H with uniform β, $c_G = n^{-1}|\{j : R_i \subseteq B_j\}|$, independent of i. On configuration models, $c_G = E_i\big[|\{j : R_i \subseteq B_j\}|/n\big]$. In general c_G is not determined by ball sizes: graphs with the same degree sequence, or the same ball-size distribution, can have different c_G because the overlap pattern of balls and residuals differs. Values must be computed by enumeration.*

Proof Definition and exchange of the order of averaging; vertex-transitivity makes the inner count independent of i. □

Proposition D.2 (Necessary condition for containment) *If $R_i \subseteq B_j$ then $\mathrm{diam}_H(R_i) \le 2r$. Hence $\tilde{c} > 0$ requires some residual of diameter at most $2r$.*

Proof For $u, v \in R_i \subseteq B_j$, $d(u,v) \le d(u,j) + d(j,v) \le 2r$. □

Proposition D.3 (R2 implies one-step extinction) *Under R2,* $\{j \in R_i : R_i \subseteq B_j\} = R_i$ *for every* i*, hence* $\tilde{c} = 1$*, and* $c_G \ge n^{-2} \sum_i |R_i|$*. Uniform balls are not required. For the 1D hypergrid with* $2\beta \ge n$*,* c_G *is given by Proposition E.1.*

Proof R2 gives $R_i \subseteq B_j$ for all $j \in R_i$; the rest follows from the definitions. □

Remark D.3.1 $\tilde{c} = 1$ throughout the exact regime is a consistency requirement for any theory of minority extinction. It does not imply that $\tilde{c}$ mediates effects outside that regime; this is tested in main-text Sec. 6.

Proposition D.4 (Intersection form, equal balls) $R_i \subseteq B_j \iff B_i \cup B_j = V \iff |B_i \cap B_j| = 2\beta - n$*. In particular containment requires* $2\beta \ge n$*, and* $c_G = 0$ *if* $2\beta < n$*.*

Proof Inclusion–exclusion, $|B_i \cup B_j| = 2\beta - |B_i \cap B_j| \le n$. □

E Exact one-dimensional containment and regions

Proposition E.1 (c_G in 1D) *On the 1D hypergrid with* $\beta = \min(n, 4r+1)$*,* $c_G = \min\big(1, \max(0, 2\beta - n + 1)/n\big)$*. At* $2\beta = n$ *the residual is itself a ball and* $c_G = 1/n$*.*

Proof The residual is an arc of $n - \beta$ nodes. For $\beta < n$, β is odd, and a ball arc of length β contains the residual if and only if its unique centre lies in an interval of $2\beta - n + 1$ positions, when this number is positive. Divide by the n possible centres. For $\beta = n$ residuals are empty and $c_G = 1$. At $2\beta = n$ the count is 1. □

Proposition E.2 ($\tilde{c}$ in 1D) *On the 1D hypergrid,* $\tilde{c} = 0$ *if* $2\beta < n$*;* $\tilde{c} = (2\beta - n + 1)/(n - \beta)$ *if* $2\beta \ge n$ *and* $3\beta < 2n - 1$ *(equal to* $1/\beta$ *at* $2\beta = n$*); and* $\tilde{c} = 1$ *if* $3\beta \ge 2n - 1$*. The function jumps from* 0 *to* $1/\beta$ *at* $2\beta = n$ *and is continuous at* $3\beta = 2n - 1$*.*

Proof The admissible centres j with $R \subseteq B_j$ form an interval of $2\beta - n + 1$ positions; intersect it with R, of $n - \beta$ nodes. If $2\beta < n$, no ball contains R. If $2\beta = n$, then $|B_j| = |R| = \beta$ and containment forces $B_j = R$, whose unique centre lies in R: exactly one admissible firing node. If $2\beta > n$ and $3\beta < 2n - 1$, the admissible interval has length $2\beta - n + 1 < n - \beta$ and lies inside R. If $3\beta \ge 2n - 1$ it covers R. At $3\beta = 2n - 1$ the middle expression equals 1. □

Remark E.2.1 The denominator is the residual size $n - \beta$, and the $+1$ is a finite-size term; neither is an asymptotic approximation.

Proposition E.3 (Three dynamical regions) *Region I:* $3\beta \geq 2n-1$ *(R1 and R2 hold, the band applies,* $\tilde{c}=1$*). Region II:* $n/2 < \beta < (2n-1)/3$ *(R1 holds and R2 fails;* $0 < \tilde{c} < 1$*). For odd* n *Region II includes* $\beta = (n+1)/2$*, the smallest ball size satisfying R1. Boundary case:* $2\beta = n$ *(*$c_G = 1/n$*,* $\tilde{c} = 1/\beta$*). Region III:* $2\beta < n$ *(R1 fails,* $\tilde{c}=0$*). Asymptotically in* q *these are* $q \geq 2/3$*,* $1/2 < q < 2/3$*, and* $q \leq 1/2$*.*

Proof Combine Proposition B.2 with Propositions E.1–E.2. □

Validation

The formulas of Propositions E.1–E.2 were compared with exhaustive enumeration of all ball–residual pairs (explicit set inclusion, rational arithmetic) for every $n \in [12, 40]$ and every radius, 192 cases in total, including the boundary case $(n, r) = (18, 2)$, where $c_G = 1/18$ and $\tilde{c} = 1/9$. All cases agree exactly (`tests/independent/containment_bruteforce.py`).

F Protocols, validation and methodological notes

Proposition F.1 (Newborn identity) *Let* $Q = 2$*. From a monochrome state, a fire at* i *creates the minority* R_i*. Conditional on the next event being a fire of a node of the minority, the firing node is uniform in* R_i *and the target is the majority colour; the minority is erased if and only if* $R_i \subseteq B_j$*. The conditional erasure probability is* $|\{j \in R_i : R_i \subseteq B_j\}|/|R_i|$*, and averaging over* i *with weights* $|R_i|$ *gives* $\tilde{c}$*.*

Proof Direct from the update rule with $Q = 2$. □

Remark F.1.1 The Monte Carlo estimate of the conditional erasure probability agrees with the enumerated $\tilde{c}$ within binomial error on native 1D balls, e.g. $n = 15$, $r = 2$: 29 489 trials, 0.6664 ± 0.0027 against $\tilde{c} = 2/3$; $n = 25$, $r = 3$: 19 535 trials, 0.1666 ± 0.0027 against $\tilde{c} = 1/6$ (`validation_results/T6.json`). On the truncated interaction sets of Protocol F.4 it agrees in all 18 cells with $0 < \tilde{c} < 1$ (largest $|z| = 1.85$; 4×10^5 fires per cell), records no erasure where $\tilde{c} = 0$, and records only erasures where $\tilde{c} = 1$ (`data/processed/newborn_test_v2_corrected.csv`). A birth is recorded when a fire from a monochrome state creates a minority, and an erasure when the next fire of a node of that minority removes it.

Proposition F.2 (Containment by intersection counting) *For equal balls,* $R_i \subseteq B_j \iff \mathrm{popcount}(B_i \,\&\, B_j) = 2\beta - n$*. Counting such pairs, with* $c_G = 0$ *when* $2\beta < n$*, computes* c_G *exactly in* $O(n^2)$ *word operations per pair.*

Proof Proposition D.4. □

Protocol F.3 (Exact solvers and simulation checks) *(i) Exact chain, reduced solver* (`tests/gvm_validation_suite.py`, task T5). States are orbits of reachable minority sets under D_n and colour exchange; the stationary distribution is computed by lazy power iteration from

the monochrome orbit, with L^1 residual $\leq 1.4 \times 10^{-14}$. This solver produced the values in Table S2.

(ii) Exact chain, independent cross-check (`tests/independent/exact_chain_independent.py`). Written separately from (i): states are full two-colour configurations with no symmetry reduction; transitions are generated literally from the update rule; irreducibility of the reachable set is checked; the stationary distribution is obtained by a sparse direct solve ($n \leq 21$) or by lazy power iteration to an L^1 change below 10^{-16} ($n = 23, 25$). In addition, the chain lumped under D_n and colour exchange is solved in exact rational arithmetic (FLINT, Dixon's algorithm), and the solution is verified by exact substitution. Results are in Table S3. The repository solver `src/exact/exact_chain_1d.py` gives the same values (`results/tables/exact_b_crosscheck.csv`).

(iii) Simulation checks (`tests/independent/sim_independent.py`). A continuous-time simulator written independently of the production simulator, 10 replicas of 2×10^6 fires per case, standard errors across replicas. Native 1D balls for the six Region-II cases of Table S2: largest $|M_{\text{sim}} - M_{\text{exact}}| = 4.6 \times 10^{-5}$, largest $|z| = 1.6$. Region I, native balls with R1 and R2 verified by breadth-first search (1D: $n = 400$, $r = 75$; $n = 401$, $r = 70$; 2D: $L = 12$, $r = 4$; $L = 30$, $r = 10$): largest $|M_{\text{sim}} - f(q)| = 2.1 \times 10^{-5}$ (largest $|z| = 0.8$); cycle statistics of Theorem C.8: largest $|z| = 1.9$ for $E[K]$ and 1.9 for CV_T^2. Full records: `results/validation/sim_independent.json`.

(iv) Newborn identity: Remark F.1.1.

Protocol F.4 (Exploratory regression test of the one-step containment hypothesis) *Design.* Families: 1D cycle; 2D torus with king (8-neighbour) adjacency; configuration model with degree sequence $k = 6$, drawn as a multigraph and projected to a simple graph (self-loops removed, multi-edges merged), so the result is not exactly 6-regular; Watts–Strogatz with $k = 6$, $p = 0.1$. Sizes $n \in \{100, 144, 400\}$; coverages $q \in \{0.55, 0.60, 0.65, 0.70, 0.75, 0.80\}$. One graph per (family, n), drawn with seed 42. Three runs (seeds 1, 2, 3) per (family, n, q).

Interaction sets. $\beta = \text{round}(qn)$; B_i consists of the β nodes nearest to i under breadth-first layering, with ties in the last layer broken at random using the run seed. The three runs of a cell therefore share the graph but not necessarily the interaction sets; $\tilde{c}$ differs between runs in 15 of the 72 cells.

Measurement. $Q = 2$, 2×10^5 fires per run, 5×10^4 discarded, M and its Monte Carlo standard error from 75 batch means of 2000 fires; c_G and $\tilde{c}$ by exact enumeration on the truncated sets.

Data. `data/processed/measurements_v2.csv`: 216 rows, one per run. The column `se` is the batch-means standard error of that run; the file is not aggregated over runs.

Model. Weighted least squares on the 216 runs with weights $1/\text{se}^2$: $M \sim q + C(\text{family}) + C(n)$ (reduced) and $M \sim q + \tilde{c} + C(\text{family}) + C(n)$ (full); γ is the coefficient of $\tilde{c}$.

Family-effect norm. $N = \left[\sum_{f<g} (\alpha_f - \alpha_g)^2\right]^{1/2}$ over all six pairs of the four family effects (1D effect = 0). N is invariant under the choice of reference family. Reduction: $(N_{\text{red}} - N_{\text{full}})/N_{\text{red}}$.

Primary bootstrap. Family-stratified cluster bootstrap: within each family, its three (family, n) clusters are resampled with replacement; within each selected cluster, the three runs are resampled with replacement inside each of the six coverage strata. Every replicate therefore contains all four families and all six coverages per cluster. $B = 2000$, seed 9999, percentile intervals.

Acceptance criteria. (a) $\gamma > 0$ with interval excluding 0; (b) reduction ≥ 0.30.

Result. $\hat{\gamma} = -0.0196$ (95% interval $[-0.0222, -0.0169]$); reduction -0.98 (interval $[-1.13, -0.85]$); fraction of replicates with reduction ≥ 0.30: 0.000. Neither criterion is met. All variants are listed in Table S4.

Status. Exploratory: the design was modified after a first analysis of the data (Note F.5).

Limitations. (1) $\tilde{c}$ describes only the first minority fire after a minority is created from a monochrome state. (2) It does not describe shrunken or fragmented minorities. (3) Within families, $\tilde{c}$ varies almost only in 1D (96.6% of the within-family sum of squares), so γ is weakly identified. (4) Only 12 clusters, three per family, each a single graph realisation; intervals do not describe graph-ensemble uncertainty. (5) Coverage enters linearly in the primary model; categorical coverage and (n, q)-cell fixed effects are reported as sensitivity analyses. (6) A failed criterion does not identify an alternative mechanism.

Note F.5 (Analysis record and sensitivity analyses) Only the acceptance criteria of Protocol F.4 were fixed before the data were analysed. The initial, pre-specified analysis resampled the 216 runs independently and measured family separation by the norm of the three family coefficients relative to the 1D reference. An intermediate analysis introduced a two-level cluster bootstrap and measured family separation by the norm of the three contrasts among the non-reference families (2D, WS, configuration). The primary analysis of Protocol F.4 replaced both choices for the reasons below; every item is reproduced by `src/analysis/reanalysis_v2.py`.

1. Point estimate. The full-sample estimates on the released data are -0.0191 without $C(n)$ and -0.0196 with $C(n)$, as stated in the protocol. The intermediate analysis used the specification without $C(n)$ and summarised the bootstrap distribution by its mean (-0.0165), which differs from the full-sample estimate. All point estimates in the main text and in Table S4 are full-sample fits.

2. Family statistic. The three-contrast norm omits all contrasts involving 1D, the only family in which $\tilde{c}$ varies appreciably, and it is not reference-invariant: choosing a different reference changes which three contrasts enter. With all six contrasts, the reduction is -0.98; the three-contrast statistic gives -0.047. The large negative value of the initial single-reference statistic (-0.865) was therefore not an artifact of the reference choice: adding $\tilde{c}$ does increase the separation of the 1D family from the others.

3. Bootstrap. The unstratified two-level scheme resamples the 12 clusters without stratification, so 61 of 500 replicates contained no cluster of at least one family (the family effect is then undefined), and it resamples the 18 runs of a cluster without regard to coverage, changing the coverage composition of each replicate. All replicates with $\hat{\gamma} > 0$ under that scheme lack at least one family; among the replicates that contain all four families, none has $\hat{\gamma} > 0$. The wide interval of that scheme, $[-0.0230, +0.0045]$, therefore reflects replicates in which γ is barely identified, not cluster-level uncertainty. The primary scheme resamples clusters within families and runs within coverage strata, and gives $[-0.0222, -0.0169]$.

4. Standard errors. The data file holds one row per run, each with its own batch-means standard error, so no aggregation of standard errors enters the fit. As a check on these standard errors, the between-run standard deviation within a cell divided by the mean batch-means standard error has median 0.832 over the 72 cells; if the standard errors are correct, the expected median for three runs is $\sqrt{\ln 2} \approx 0.833$. Fits to the 72 cell means, with standard errors combining within- and between-run variation, give $\hat{\gamma} = -0.0188$.

5. Dependence. Runs that share a (family, n) cluster share a graph, so resampling them independently does not represent the design. In these data it makes little difference: the iid interval is $[-0.0223, -0.0170]$, close to the primary interval. Resampling runs only within cells, which reflects Monte Carlo noise alone, gives $[-0.0200, -0.0192]$. The wide interval of the

unstratified scheme is therefore due to the absence of stratification (item 3), not to cluster dependence.

6. Coverage variable. The regressor q is the nominal grid value; the realised coverage is β/n with $\beta = \mathrm{round}(qn)$, which differs by at most $0.5/n$. Using β/n gives $\hat{\gamma} = -0.0201$.

Protocol F.6 (Extended Region-II solutions and annealed analogue) *(i) Quenched chain* (`scripts/region2_exact.c`). Written independently of the solvers of Protocol F.3. States are the two-colour configurations reachable from the monochrome state by breadth-first search, identified under colour exchange only (no dihedral reduction). Every node fires at rate 1, so the total jump rate is n in every state and the stationary law of the continuous-time chain equals that of the jump chain $P = n^{-1}\sum_i T_i$, where T_i is the deterministic update of a fire at i. The stationary law is computed by power iteration of $(I+P)/2$ from the uniform distribution, with renormalisation at every step, until the L^1 change falls below 10^{-12}, followed by 3000 further iterations; the reported residual is $\|\pi P - \pi\|_1$, below 5×10^{-13} in every case. For the six cases of Table S2 the code reproduces the reachable-set sizes and the exact rationals to 10^{-12}.

(ii) Annealed analogue (`scripts/annealed_baseline.py`). At every fire of i the interaction set is $\{i\}$ together with $\beta - 1$ nodes drawn uniformly without replacement from $V \setminus \{i\}$. If i has a colour shared by m nodes, the number of moved nodes is $1 + h$ with h hypergeometric (population $n-1$, $m-1$ successes, $\beta - 1$ draws). The dynamics is a Markov chain on the count $k \in \{0, \ldots, n\}$ of one colour; its stationary law is obtained by a direct linear solve, with residual below 2×10^{-13} for all sizes used.

(iii) Table (`scripts/make_region2_table.py`) assembles main-text Table 2 from (i) and (ii).

Validation ledger

Table S1 lists every numerical check cited in the main text and its evidence file.

G Conjectures and open problems

Conjecture G.1 *In Region II, $E[M] \le f(q)$ for all n and admissible r.*

Evidence. Exact rational solutions for the six odd $n \le 25$ in Region II (Table S2), with deviations from -0.0021 to -0.0184. The rationals $43/57$, $65/84$, $149/201$, $265/339$, $72/95$, $319/433$ show no evident common form. The independent solver of Protocol F.6 extends the evidence to all 17 Region-II cases with $n \le 33$, odd and even (main text, Table 2), with deviations from -0.0009 to -0.0195, increasing monotonically with q. No coupling or monotonicity proof is known.

Conjecture G.2 (Scaling near threshold) *A geometric scaling form governs M near the threshold on d-dimensional hypergrids. Untested; no exponents are proposed.*

Heuristic G.3 (One-step containment as mediator; not supported) *The hypothesis that $\tilde{c}$ mediates the family dependence of M at intermediate coverage was to be accepted only if both criteria of Protocol F.4 held. Neither holds in the tested design.*

Table S1 Validation ledger. All entries were executed for this version; files are relative to the repository root.

Check	Outcome	Evidence
1D containment formulas vs enumeration	exact agreement, 192 cases	`results/validation/containment_bruteforce.json`
Exact chain, Region II (6 cases), two independent solvers	agree; exact rationals certified	`results/validation/exact_independent.json`; `validation_results/T5.json`
Exact chain, Region I (band)	equals $f(q)$	`validation_results/T1.json`
Conjecture G.1, six cases	no violation (exact)	Table S2
Simulator vs exact chain, Region II	$\lvert\Delta\rvert \le 4.6 \times 10^{-5}$	`results/validation/sim_independent.json`
Simulator vs band, Region I (1D, 2D)	$\lvert\Delta\rvert \le 2.1 \times 10^{-5}$	`results/validation/sim_independent.json`; `validation_results/T3.json,T4.json`
Cycle law, $E[K]$ and CV_T^2	$\lvert z\rvert \le 1.9$, 1.9	`results/validation/sim_independent.json`
Newborn identity (native 1D; truncated sets)	within binomial error; $\lvert z\rvert \le 1.85$	`validation_results/T6.json`; `data/processed/newborn_test_v2_corrected.csv`
Regression test (Protocol F.4)	executed; criteria not met	`results/tables/reanalysis/`
Region II, all cases $n \le 33$; annealed analogue (Protocol F.6)	all below $f(q)$; annealed below quenched	`scripts/region2_results.txt`

Open problem G.4 (Configuration–2D gap) *Candidate explanations of the difference between the configuration-model and 2D families include residual ball-size heterogeneity (partly removed by truncation), expansion properties, and fractional-radius effects. Truncation to equal β (Protocol F.4) does not remove the difference.*

Heuristic G.5 (Fragmentation) *Below R2, minorities may fragment. No low-dimensional closed Markov description in terms of shape variables is known.*

Limitations and scope

1. *Regimes without an exact characterisation.* The band (Theorem C.6) is proved only under R1∧R2 with equal ball sizes. Regions II and III have no closed-form stationary characterisation; Conjecture G.1 is supported by six exact rational cases and by floating-point solutions of all Region-II cases with $n \le 33$.
2. *Statements with proof sketches.* Claims B.4 and B.5 are not used as hypotheses of any result.
3. *Single graph realisation.* Protocol F.4 uses one graph per (family, n). Family offsets are properties of these graphs, not of the graph ensembles.

Table S2 Exact chain in Region II of the 1D hypergrid (all odd $n \leq 25$). "Reachable": minority sets reachable from the monochrome state (the reachable set is one irreducible class in every case); "orbits": classes under D_n and colour exchange. $E[M]$ is the exact rational value from the lumped chain.

n	r	β	q	c_G	$\tilde{c}$	$E[M]$	$f(q)$	reachable	orbits
9	1	5	0.5556	2/9	1/2	43/57	13/17	73	8
15	2	9	0.6000	4/15	2/3	65/84	7/9	481	24
17	2	9	0.5294	2/17	1/4	149/201	25/33	2177	80
21	3	13	0.6190	2/7	3/4	265/339	29/37	2689	80
23	3	13	0.5652	4/23	2/5	72/95	33/43	11777	288
25	3	13	0.5200	2/25	1/6	319/433	37/49	51201	1088

Table S3 Independent cross-check of Table S2: full two-colour configurations without symmetry reduction, sparse direct solve. "Config.": reachable configurations (twice the number of reachable minority sets). Last column: floating-point value minus the exact rational of Table S2.

n	r	config.	$E[M]$ (floating point)	L^1 residual	difference
9	1	146	0.7543859649	1.4×10^{-15}	-1.1×10^{-16}
15	2	962	0.7738095238	1.3×10^{-14}	1.1×10^{-16}
17	2	4354	0.7412935323	2.7×10^{-14}	1.2×10^{-14}
21	3	5378	0.7817109145	4.1×10^{-14}	5.7×10^{-15}
23	3	23554	0.7578947368	9.6×10^{-16}	-2.1×10^{-15}
25	3	102402	0.7367205543	9.4×10^{-15}	2.1×10^{-15}

4. *Identification.* $\tilde{c}$ varies almost only within the 1D family, and coverage enters linearly in the primary model.
5. *Mechanism.* The Region-II observation $M_{\text{exact}} < f(q)$ is consistent with multi-step minority survival, but no result here identifies that mechanism (Heuristic G.5).
6. *Equal balls.* The band requires $\beta_i \equiv \beta$; Proposition C.1 and Proposition D.3 do not.
7. *2-section reduction.* The dynamics depends on the hypergraph only through its 2-section graph (main-text Remark 1); no higher-order mechanism beyond the pairwise projection is claimed.
8. *Few clusters.* Twelve clusters give wide intervals, and their width depends on the resampling scheme (Table S4).

H Numerical tables

Reproducibility

Code and data: https://github.com/phmldtngyn/Group-flip-consensus-on-hypergraphs. The analysis outputs cited here are in `results/tables/reanalysis/` and `results/validation/`; the commands are listed in the repository `README.md`. A preliminary

Table S4 Regression test, Protocol F.4: all specifications and resampling schemes. "Point": full-sample estimate. Intervals: bootstrap percentile 95%. Reduction: six-contrast family-norm reduction (negative means increased separation). "Missing": replicates lacking at least one family. Rows marked † use the specification of the intermediate analysis (no $C(n)$, $B = 500$); "primary": the specification of Protocol F.4.

Specification	Bootstrap	B	$\hat{\gamma}$ (point)	γ 95%	reduction 95%	missing
lin. q, no $C(n)^{\dagger}$	unstratified cluster	500	-0.0191	$[-0.0230, +0.0045]$	$[-1.269, -0.003]$	61
lin. q, no $C(n)^{\dagger}$	runs iid	500	-0.0191	$[-0.0220, -0.0163]$	$[-1.251, -0.737]$	0
primary: lin. q, $C(n)$	unstratified cluster	2000	-0.0196	$[-0.0236, +0.0022]$	$[-1.365, -0.001]$	258
primary: lin. q, $C(n)$	stratified cluster	2000	-0.0196	$[-0.0222, -0.0169]$	$[-1.129, -0.849]$	0
primary: lin. q, $C(n)$	runs iid	2000	-0.0196	$[-0.0223, -0.0170]$	$[-1.249, -0.753]$	0
primary: lin. q, $C(n)$	within cells	2000	-0.0196	$[-0.0200, -0.0192]$	$[-1.021, -0.937]$	0
lin. q, no $C(n)$	stratified cluster	2000	-0.0191	$[-0.0221, -0.0165]$	$[-1.116, -0.821]$	0
categ. q, $C(n)$	stratified cluster	2000	-0.0185	$[-0.0208, -0.0161]$	$[-1.074, -0.802]$	0
(n, q)-cell FE	stratified cluster	2000	-0.0186	$[-0.0215, -0.0159]$	$[-1.119, -0.772]$	0
primary spec., OLS	—	—	-0.0181	—	point -0.84	—
primary spec., 72 cell means	—	—	-0.0188	—	point -0.94	—

measurement run on native balls, in which the coverage variable was mislabelled, was discarded before any analysis and is not used.